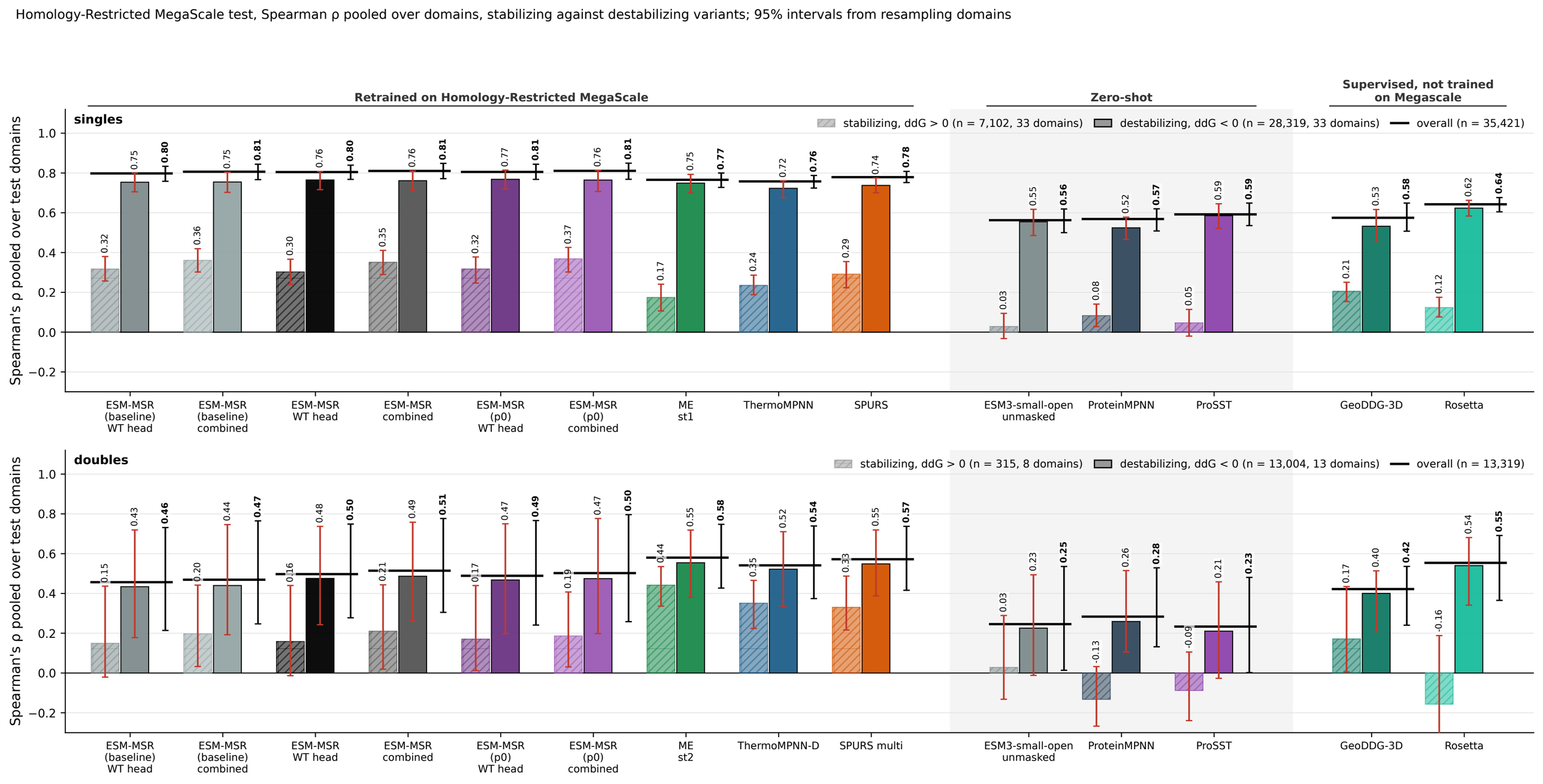
Homology-Restricted MegaScale test, Spearman ρ pooled over domains, stabilizing against destabilizing variants; 95% intervals from resampling domains
Retrained on Homology-Restricted MegaScale
Zero-shot
Supervised, not trained on Megascale
singles
stabilizing, ddG > 0 (n = 7,102, 33 domains)
destabilizing, ddG < 0 (n = 28,319, 33 domains)
overall (n = 35,421)
Spearman's ρ pooled over test domains
ESM-MSR (baseline) WT head
ESM-MSR (baseline) combined
ESM-MSR WT head
ESM-MSR combined
ESM-MSR (p0) WT head
ESM-MSR (p0) combined
ME st1
ThermoMPNN
SPURS
ESM3-small-open unmasked
ProteinMPNN
ProSST
GeoDDG-3D
Rosetta
doubles
stabilizing, ddG > 0 (n = 315, 8 domains)
destabilizing, ddG < 0 (n = 13,004, 13 domains)
overall (n = 13,319)
ME st2
ThermoMPNN-D
SPURS multi